Dhruv Pathak

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GitHub: github.com/DhruvPathak2406
HackTheBox: Hack The Box Profile

Profile

Computer Science undergraduate with foundations in software development, penetration testing, and cybersecurity, seeking internship opportunities in related domains. Experienced in building ML-driven solutions, with strong analytical, problem-solving, and teamwork skills. Eligible to work in both the United Kingdom and India.

Education

Sardar Vallabhbhai National Institute of Technology **2024 – 2028**
B.Tech in Computer Science and Engineering *CGPA: 7.28*
Rahul International School **2023**
Class 12 *Score: 87.2%*

Relevant Experience

Core - Event Lead **August 2025 – Present**
Nexus - DoCSE Cell
- Organised and executed several events, ensuring effective planning and coordination.
- Contributed to the organisation of **E-Rakshak**, a joint hackathon between Surat Police and Nexus.
- Contributed to the organisation of **Uncharted**, a joint hackathon between MongoDB and Nexus.
- Supported logistics, participant management, and technical coordination across club activities.
- Mentored **9 students** while contributing to daily operations, demonstrating leadership and teamwork.

Executive **September 2025 – August 2026**
Cutting Edge Visionaries
- Participated in the **Genesis AI/ML Bootcamp and Hackathon**, securing Runner-Up position while supporting club operations and collaborative initiatives.

Projects

BlindSpot **GitHub**
Tech: Python, NetworkX, NumPy, Pandas, Bayesian Inference, Reinforcement Learning, HTML, CSS, JavaScript
- Developed an adaptive network reconnaissance platform that models partially observable networks and prioritises discovery actions under constrained reconnaissance budgets.
- Implemented **Bayesian belief modelling** and entropy-based information gain to quantify network uncertainty and optimise host, port, service, and topology discovery.
- Integrated **contextual bandits and reinforcement learning** for adaptive reconnaissance, benchmarking performance against random and fixed-order reconnaissance strategies.
- Built an interactive web dashboard for network visualisation, reconnaissance monitoring, and performance analysis.

ForecastX (Genesis Hackathon - Runner Up) **GitHub**
Tech: Python, Pandas, NumPy, Scikit-learn, XGBoost, yFinance, Matplotlib
- Built an AI-based stock price prediction system for long-term trading using machine learning and financial data analysis.
- Applied an ensemble of **Random Forest** and **XGBoost** to improve prediction accuracy and robustness.
- Developed a pipeline for data extraction, preprocessing, model training, and visualisation using financial market data.

Cognis Sentri **GitHub**
Tech: Python, OpenCV, DeepFace, NumPy, Scikit-learn, HTML, CSS, JavaScript
- Developed an AI-powered surveillance system for real-time face recognition and threat detection using computer vision.
- Built a recognition pipeline using **OpenCV** and **DeepFace** to classify known and unknown individuals.
- Integrated a web frontend and simulated alert systems using buzzers and LEDs for real-time response.
- Implemented reinforcement learning-based threat assessment and automated alert mechanisms.

Technical Skills

Languages: Python, C++, C, SQL
Developer Tools: Git, GitHub, VS Code, Linux
Security Tools: Nmap, Metasploit, Wireshark, FFUF, Hydra, John the Ripper, Hashcat
Frameworks and Libraries: Scikit-learn, OpenCV, DeepFace, Burp Suite
Databases: MySQL, MSSQL
Soft Skills: Analytical Thinking, Teamwork, Problem Solving, Adaptability
Coursework: Computer Networks, Object-Oriented Programming, Database Management Systems, Data Structures and Algorithms
Areas of Interest: Red Teaming, Penetration Testing, Network Security, VAPT, Artificial Intelligence, Machine Learning

Achievements

- Completed the **Certified Penetration Testing Specialist (CPTS) Job-Role Path** by Hack The Box, covering network enumeration, privilege escalation, pivoting, and penetration testing methodologies.
- Completed the **Penetration Testing-1 (PT-1) Certification Path** by TryHackMe, successfully compromising 50+ machines.
- Completed the **Google Cybersecurity Professional Certificate** (Coursera), covering networking, Linux, SQL, SIEM, threat detection, and incident response.
- Secured Runner-Up position at the **Genesis AI/ML Bootcamp** with **ForecastX**.
- Completed certifications in **DSA, Python (Udemy), Machine Learning (University of Michigan)**, and **Market Analysis (University of Geneva)**.